

Maintenance Schedule

Diesel Engine

EPA/CARB

10V1600T60

12V1600T60

Application Group 4B

MSE50220/00E



Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
10V1600T60	57 kW/cyl.	4B, Continuous operation, variable
12V1600T60	57 kW/cyl.	4B, Continuous operation, variable

Table 1: Applicability

Table of Contents

1	Maintenance Schedule		
1.1	Preface	5	
1.2	Allocation matrix application group 4B	6	
1.3	Maintenance schedule matrix 21,000 h	8	
1.4	Maintenance tasks 21,000 h	9	
1.5	Maintenance schedule matrix 18,000 h	10	
1.6	Maintenance tasks 18,000 h	11	
1.7	Maintenance schedule matrix 15,000 h	12	
1.8	Maintenance tasks 15,000 h	13	
1.9	Maintenance schedule matrix 12,000 h	14	
1.10	Maintenance tasks 12,000 h	15	
1.11	Maintenance schedule matrix 9,000 h	16	
1.12	Maintenance tasks 9,000 h	17	

1 Maintenance Schedule

1.1 Preface

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

Information about the latest maintenance schedule is available at: <http://www.mtu-online.com>.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emission specifications. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-relevant components may only be maintained, exchanged or repaired if the components used for this purpose are approved by MTU or equivalent components. Emission-relevant components are for instance control units, data records, sensors, injectors, exhaust flaps and all components of exhaust aftertreatment systems.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals and the scope of required maintenance activities are the result of operational experience and therefore represent recommendations. Severe operating and ambient conditions may require additional maintenance work and/or modification of the maintenance intervals.

Maintenance activities are specified with intervals based on hours in operation and time limits. Whichever value is reached first shall apply.

The individual maintenance intervals are classified in accordance with the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

For change intervals for fluids and lubricants as well as for preservation specifications, refer to the relevant instructions of the component manufacturers and the MTU Fluids and Lubricants Specifications. The latest version for drive systems is available online at: <http://www.mtu-online.com>, and for power generation systems at: <http://www.mtuonsiteenergy.com>.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

1.2 Allocation matrix application group 4B

Load profile		10/12V1600T60						
Load %	Time %	Load factor %	Load indicator %	0 to 57 kW/cyl.				
110 ₍₁₀₀₋₁₁₀₎	—	0 to 31	≤ 17	21,000	—	—	—	Standard
100 ₍₉₀₋₁₀₀₎	12							
90 ₍₈₀₋₉₀₎	12							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	16							
Idle Speed ₍₀₋₅₎	60							
110 ₍₁₀₀₋₁₁₀₎	—	32 to 35	≤ 19	18,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	17							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	13							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle Speed ₍₀₋₅₎	55							
110 ₍₁₀₀₋₁₁₀₎	—	36 to 45	≤ 32	15,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	26							
90 ₍₈₀₋₉₀₎	15							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	12							
Idle Speed ₍₀₋₅₎	47							
110 ₍₁₀₀₋₁₁₀₎	—	46 to 55	≤ 41	15,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	35							
90 ₍₈₀₋₉₀₎	15							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle Speed ₍₀₋₅₎	35							

Table 2: Allocation matrix application group 4B

TIM-ID: 000006684 - 001

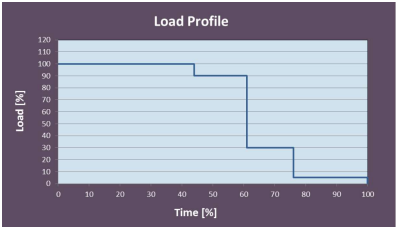
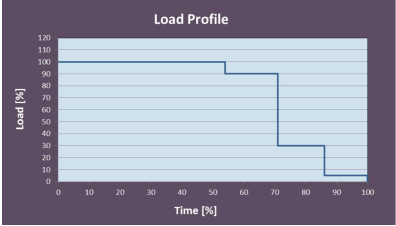
10/12V1600T60								
Load profile		Load factor %	Load indicator %	0 to 57 kW/cyl				
Load %	Time %							
110 ₍₁₀₀₋₁₁₀₎	—	56 to 65	≤ 51	12,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	44							
90 ₍₈₀₋₉₀₎	17							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle Speed ₍₀₋₅₎	24							
110 ₍₁₀₀₋₁₁₀₎	—	66 to 75	≤ 61	9,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	54							
90 ₍₈₀₋₉₀₎	17							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle Speed ₍₀₋₅₎	14							

Table 3: Allocation matrix application group 4B